Practice workbook · Solving an equation with your graph and a straight line

Rearranging a polynomial, reciprocal or trigonometric equation so that one side is the plotted curve, drawing the matching straight line and reading off the roots.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-GRA-08

1. Question 8 · part c · 5 marks

2021 NOV · Paper 02

Worked-example retry · Attempt part c only.

8 (a) Complete the table of values for $y = 2x + \frac{3}{x^2} - 3$ giving your answers to 2 decimal places where appropriate.

x	0.5	0.75	1	1.5	2	3	4	5
y	10		2				5.19	7.12

(2)

(b) On the grid opposite, draw the graph of $y = 2x + \frac{3}{x^2} - 3$ for $0.5 \leqslant x \leqslant 5$

(2)

(c) By drawing a suitable straight line on the grid, obtain estimates, to one decimal place, of the roots of the equation $4x^3 - 10x^2 + 3 = 0$ for $0.5 \leqslant x \leqslant 5$

(5)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
OLevels.com SCHOOL

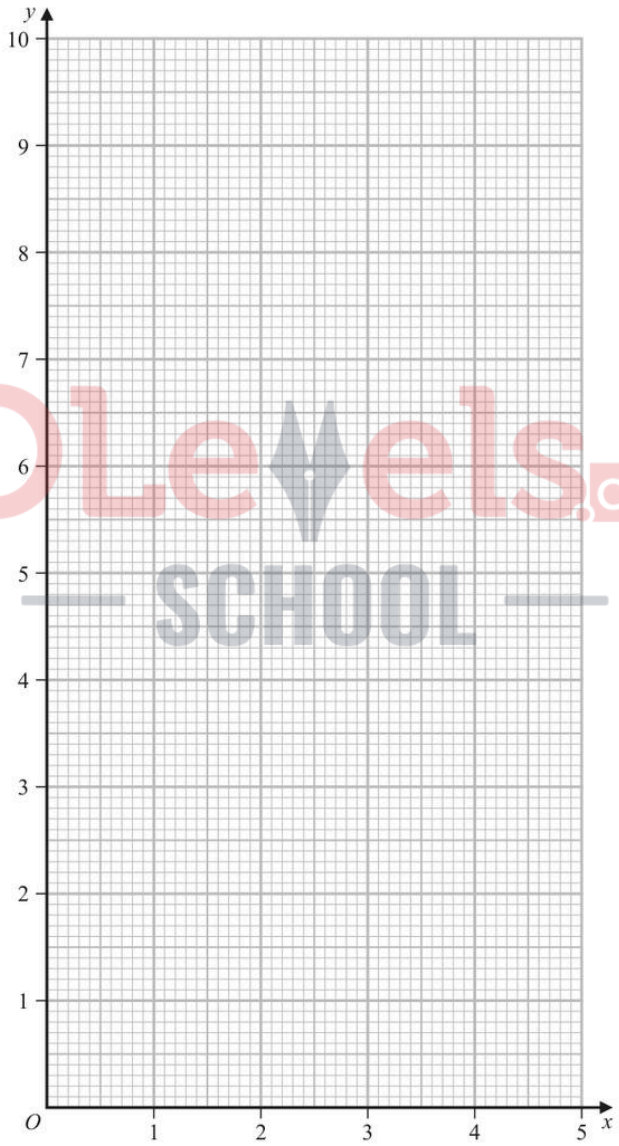
P 6 6 0 2 7 R R A 0 2 2 3 6

1. Question 8 · part c · 5 marks

2021 NOV · Paper 02

Worked-example retry · Attempt part c only.

Question 8 continued



Turn over for a spare grid if you need to redraw your graph.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

P 6 6 0 2 7 R R A 0 2 3 3 6

23

Turn over ▶

1. Question 8 · part c · 5 marks

2021 NOV · Paper 02

Worked-example retry · Attempt part c only.

Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

24

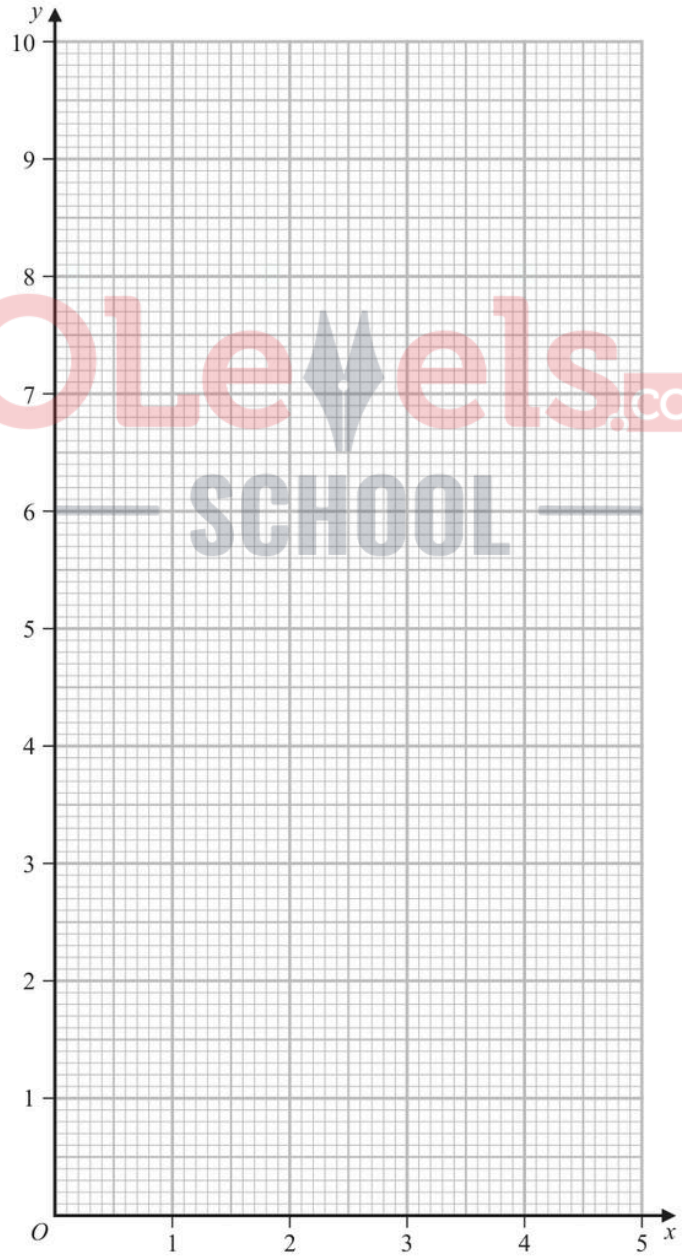
1. Question 8 · part c · 5 marks

2021 NOV · Paper 02

Worked-example retry · Attempt part c only.

Question 8 continued

Only use this grid if you need to redraw your graph.



DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(Total for Question 8 is 9 marks)

1. Question 8 · part c · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 8 · part c · 3 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

8 (a) Complete the table of values for $y = 2 + \ln(2x + 1)$ giving your answers to 2 decimal places.

(2)

x	0	0.25	0.5	1	1.5	2	3
y	2			3.10	3.39	3.61	

(b) On the grid opposite, draw the graph of $y = 2 + \ln(2x + 1)$ for $0 \leqslant x \leqslant 3$

(2)

(c) By drawing an appropriate straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation $\ln(2x + 1) = 3x - 4$ in the interval $0 \leqslant x \leqslant 3$

(3)

(d) By drawing an appropriate straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation $e^{(6-x)} - (2x + 1)^2 = 0$ in the interval $0 \leqslant x \leqslant 3$

(4)

OLevels.com SCHOOL

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

20

P 6 1 8 8 2 A 0 2 0 3 6

2. Question 8 · part c · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 4 · part a · 3 marks

2020 NOV · Paper 01

Further practice · Attempt part a only.

4

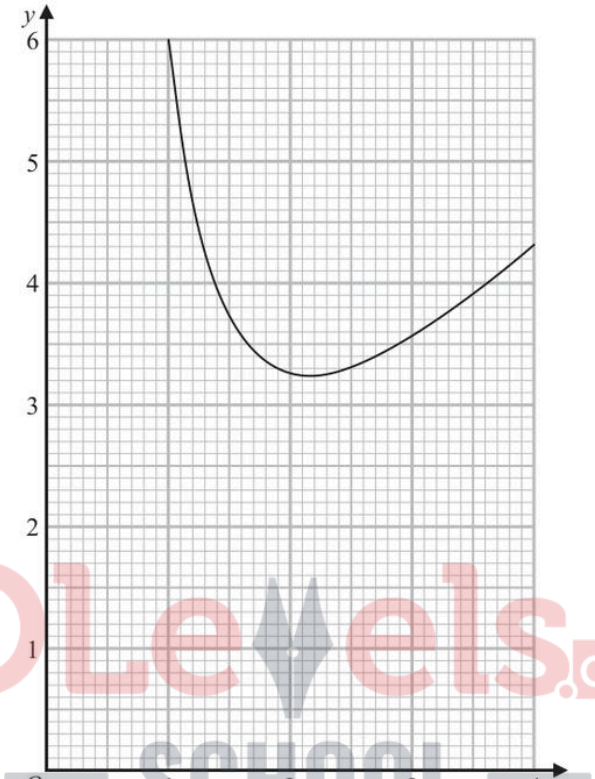


Figure 2

Figure 2 shows the graph of $y = x + \frac{5}{x^2}$ for $1 \leqslant x \leqslant 4$ drawn on a grid.

(a) By drawing a suitable straight line on the grid, obtain estimates, to one decimal place, for the roots of the equation

$$x^3 - 4x^2 + 5 = 0$$

in the interval $1 \leqslant x \leqslant 4$

(3)

(b) By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, for the root of the equation

$$x^3 - x^2 - 5 = 0$$

in the interval $1 \leqslant x \leqslant 4$

(4)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

3. Question 4 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 4 · part b · 4 marks

2020 NOV · Paper 01

Further practice · Attempt part b only.

4

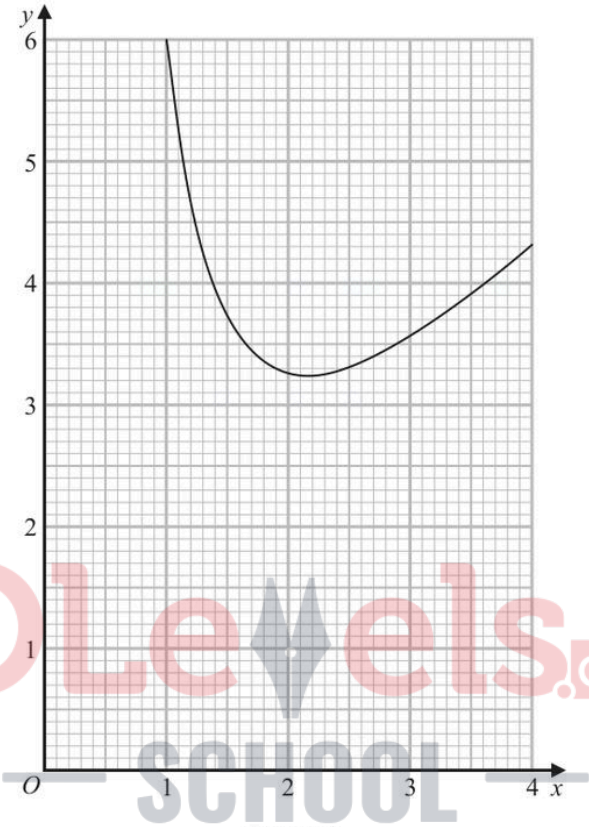


Figure 2

Figure 2 shows the graph of $y = x + \frac{5}{x^2}$ for $1 \leqslant x \leqslant 4$ drawn on a grid.

(a) By drawing a suitable straight line on the grid, obtain estimates, to one decimal place, for the roots of the equation

$$x^3 - 4x^2 + 5 = 0$$

in the interval $1 \leqslant x \leqslant 4$

(3)

(b) By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, for the root of the equation

$$x^3 - x^2 - 5 = 0$$

in the interval $1 \leqslant x \leqslant 4$

(4)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

4. Question 4 · part b · 4 marks

2020 NOV · Paper 01

Further practice · Attempt part b only.

Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 4 is 7 marks)

4. Question 4 · part b · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 4 · part c · 2 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part c only.

4 (a) Complete the table of values for $y = 2x + 1 + \frac{2}{x^2}$

Give your answers to 2 decimal places where appropriate.

x	0.5	1	1.5	2	2.5	3	3.5
y		5			6.32		8.16

(2)

(b) On the grid opposite, draw the graph of $y = 2x + 1 + \frac{2}{x^2}$ for $0.5 \leqslant x \leqslant 3.5$

(2)

(c) Use your graph to obtain estimates, to 1 decimal place, of the roots of the equation

$$2x + \frac{2}{x^2} = 7 \quad \text{in the interval } 0.5 \leqslant x \leqslant 3.5$$

(2)

(d) By drawing a suitable straight line on the grid, obtain estimates, to 1 decimal place, of the roots of the equation

$$\frac{3x}{2} + \frac{2}{x^2} = 5 \quad \text{in the interval } 0.5 \leqslant x \leqslant 3.5$$

(5)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

5. Question 4 · part c · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 4 · part d · 5 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part d only.

4 (a) Complete the table of values for $y = 2x + 1 + \frac{2}{x^2}$

Give your answers to 2 decimal places where appropriate.

x	0.5	1	1.5	2	2.5	3	3.5
y		5			6.32		8.16

(2)

(b) On the grid opposite, draw the graph of $y = 2x + 1 + \frac{2}{x^2}$ for $0.5 \leqslant x \leqslant 3.5$

(2)

(c) Use your graph to obtain estimates, to 1 decimal place, of the roots of the equation

$$2x + \frac{2}{x^2} = 7 \quad \text{in the interval } 0.5 \leqslant x \leqslant 3.5$$

(2)

(d) By drawing a suitable straight line on the grid, obtain estimates, to 1 decimal place, of the roots of the equation

$$\frac{3x}{2} + \frac{2}{x^2} = 5 \quad \text{in the interval } 0.5 \leqslant x \leqslant 3.5$$

(5)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

6. Question 4 · part d · 5 marks

2020 NOV · Paper 02 (regional R)

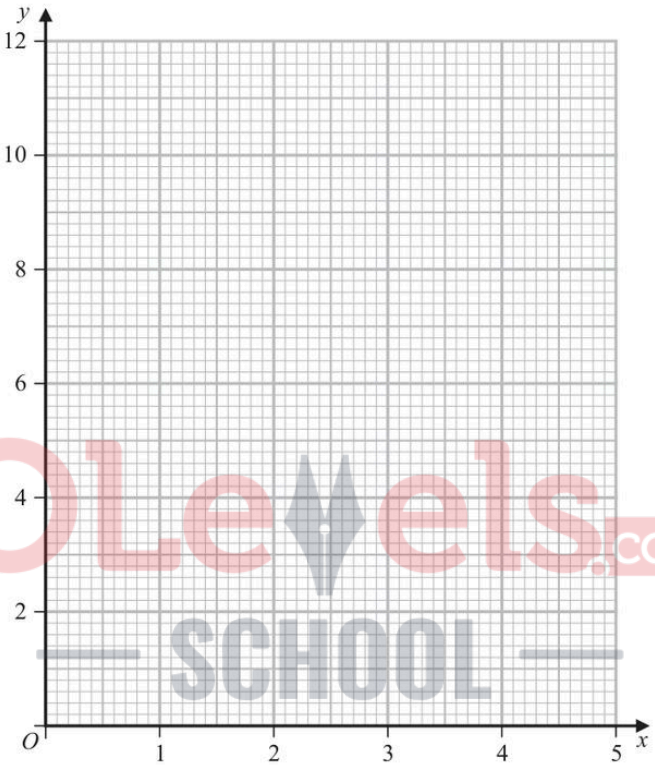
Further practice · Attempt part d only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 4 continued



OLevels.COM SCHOOL

Turn over for a spare grid if you need to redraw your graph.

P 6 2 2 8 4 A 0 9 3 2

9

Turn over ▶

6. Question 4 · part d · 5 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

10

P 6 2 2 8 4 A 0 1 0 3 2

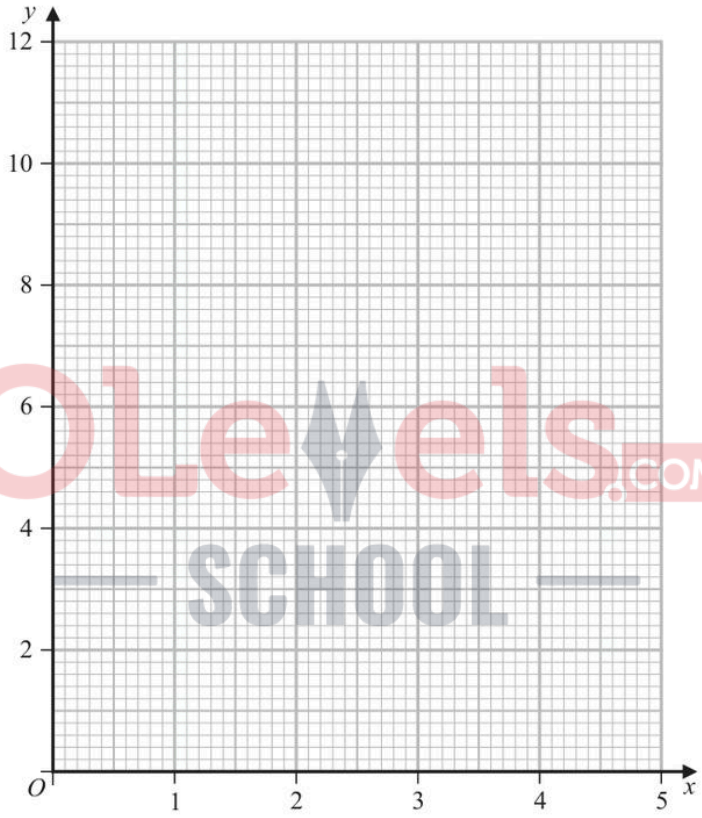
6. Question 4 · part d · 5 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 4 continued

Only use this grid if you need to redraw your graph.



DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(Total for Question 4 is 11 marks)

6. Question 4 · part d · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 8 · part d · 5 marks

2022 JAN · Paper 02

Further practice · Attempt part d only.

8 (a) Given that x is measured in radians, complete the table of values for

$$y = \sin x - 3\cos 2x - 0.5$$

Give your answers to one decimal place.

x	0	0.5	0.8	1	1.6	2	2.5	3
y			0.3	1.6			–0.8	–3.2

(2)

(b) On the grid opposite, draw the graph of $y = \sin x - 3\cos 2x - 0.5$ for $0 \leqslant x \leqslant 3$

(2)

(c) Use a formula from page 2 to show that $\cos 2A = 1 - 2\sin^2 A$

(2)

$$f(x) = 2\sin x + 12\sin^2 x - x - 5$$

(d) By drawing a suitable straight line on the grid, obtain estimates, to one decimal place, of the roots of the equation $f(x) = 0$ in the interval $0 \leqslant x \leqslant 3$

(5)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

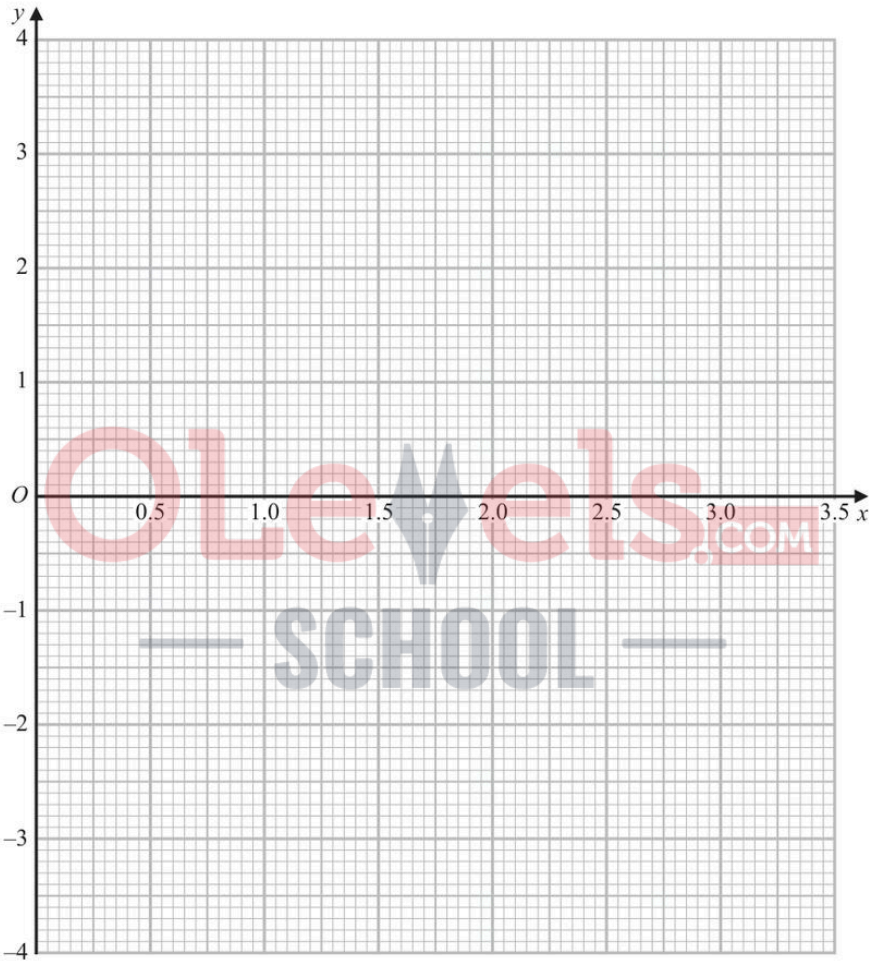
P 6 6 3 0 8 A 0 2 4 3 6

7. Question 8 · part d · 5 marks

2022 JAN · Paper 02

Further practice · Attempt part d only.

Question 8 continued



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Turn over for a spare grid if you need to redraw your graph.

7. Question 8 · part d · 5 marks

2022 JAN · Paper 02

Further practice · Attempt part d only.



Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

26

P 6 6 3 0 8 A 0 2 6 3 6

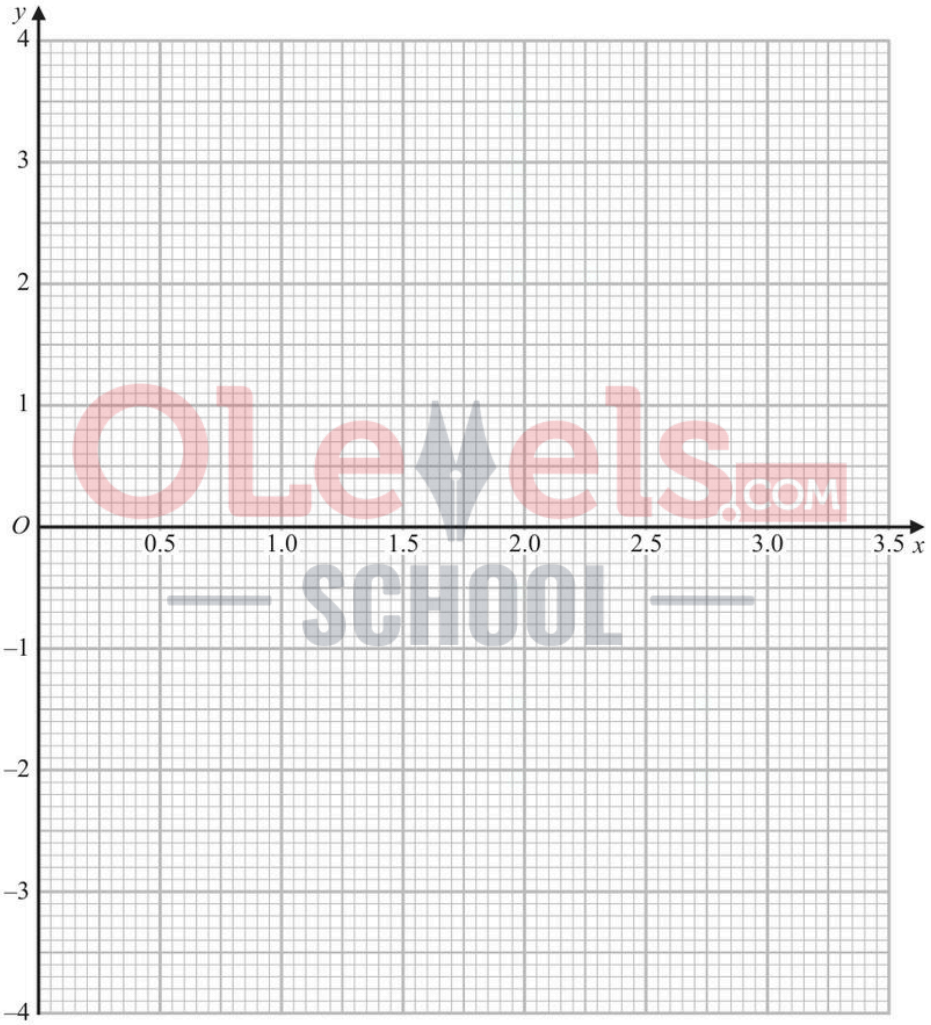
7. Question 8 · part d · 5 marks

2022 JAN · Paper 02

Further practice · Attempt part d only.

Question 8 continued

Only use this grid if you need to redraw your graph.



DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(Total for Question 8 is 11 marks)

7. Question 8 · part d · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 6 · part c · 5 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

6 (a) Complete the table of values for

$$y = x - \frac{3}{x^2}$$

giving your answers to one decimal place where appropriate.

x	0.5	1	1.5	2	3	4	5	6
y	–11.5			1.3	2.7			5.9

(2)

(b) On the grid opposite, draw the graph of $y = x - \frac{3}{x^2}$ for $0.5 \leqslant x \leqslant 6$

(2)

(c) By drawing a suitable straight line on the grid, obtain estimates, to one decimal place, of each of the two roots of the equation

$$2x^3 - 6x^2 + 3 = 0$$

in the interval $0.5 \leqslant x \leqslant 6$

(5)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

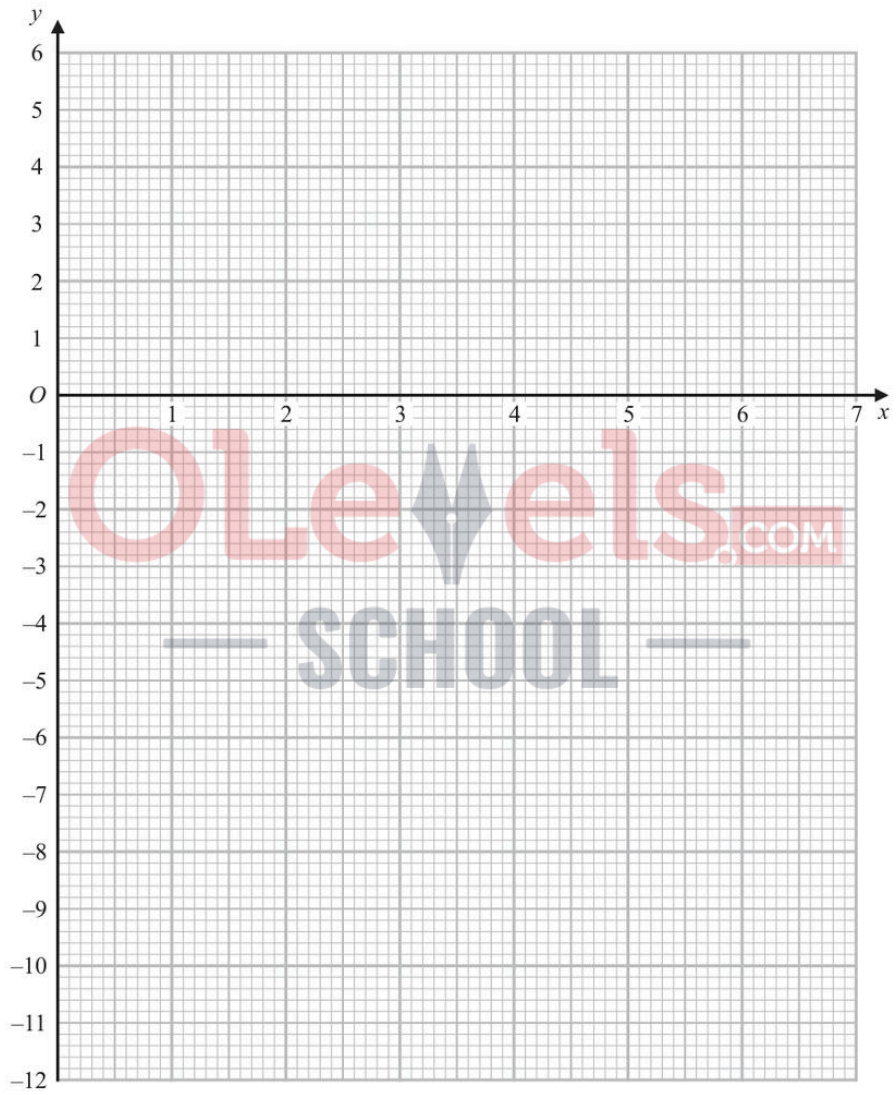
P 6 0 4 7 5 A 0 1 2 3 6

8. Question 6 · part c · 5 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 6 continued



OLevels.com SCHOOL

Turn over for a spare grid if you need to redraw your graph.

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

8. Question 6 · part c · 5 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

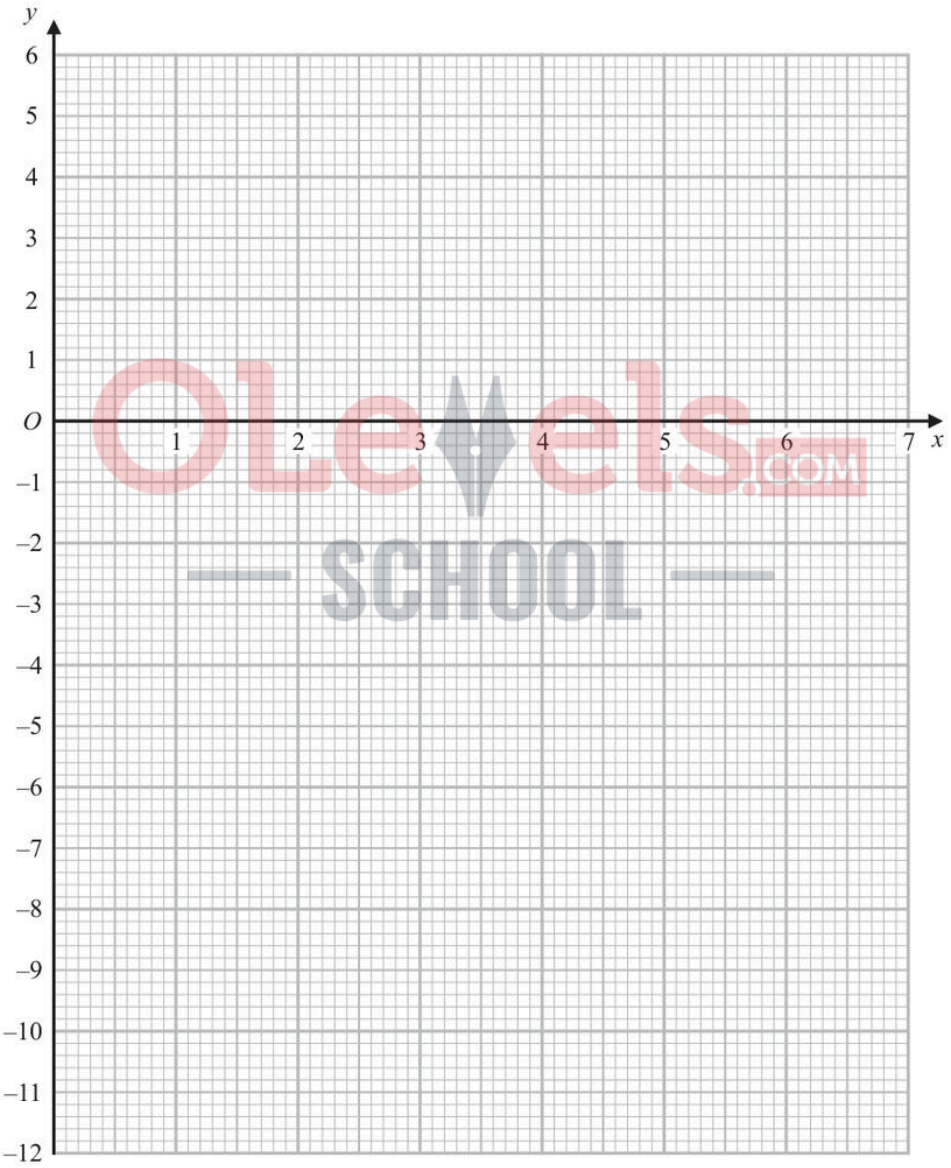
8. Question 6 · part c · 5 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 6 continued

Only use this grid if you need to redraw your graph.



DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(Total for Question 6 is 9 marks)

8. Question 6 · part c · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

9. Question 6 · part c · 3 marks

2022 JUN · Paper 01

Further practice · Attempt part c only.

6 (a) Complete the table of values for $y = 1 + 3e^{-x}$ giving your answers to 2 decimal places where appropriate.

(2)

x	0	0.25	0.5	1	1.5	2	3
y		3.34	2.82		1.67		1.15

(b) On the grid opposite, draw the graph of $y = 1 + 3e^{-x}$ for $0 \leqslant x \leqslant 3$

(2)

(c) By drawing an appropriate straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$x = e^{-x} \quad \text{in the interval } 0 \leqslant x \leqslant 3$$

(3)

(d) By drawing an appropriate straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$\ln(x-1)^3 = -3x \quad \text{in the interval } 0 \leqslant x \leqslant 3$$

(4)

OLevels.COM — SCHOOL —

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

9. Question 6 · part c · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

10. Question 3 · part whole · 5 marks

2023 JUN · Paper 01

Further practice · Attempt part whole only.

3

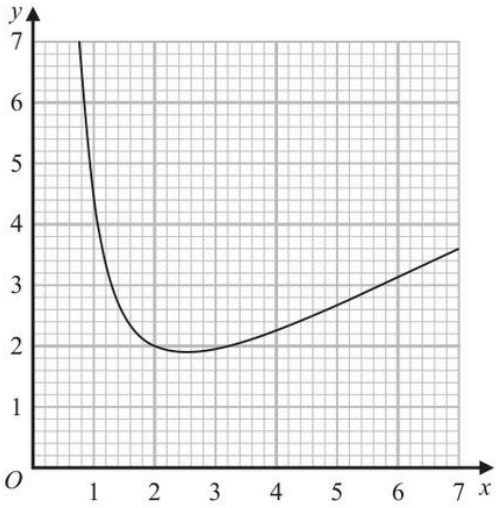


Figure 1

Figure 1 shows part of the curve with equation $y = \frac{x}{2} + \frac{4}{x^2}$ in the interval $0.8 < x < 7$

By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, of the roots of the equation $3x^3 - 12x^2 + 8 = 0$ in the interval $0.8 < x < 7$

(5)

OLevels.com — SCHOOL —

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

10. Question 3 · part whole · 5 marks

2023 JUN · Paper 01

Further practice · Attempt part whole only.

Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 3 is 5 marks)

10. Question 3 · part whole · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

11. Question 4 · part whole · 4 marks

2024 JUN · Paper 02

Further practice · Attempt part whole only.

4

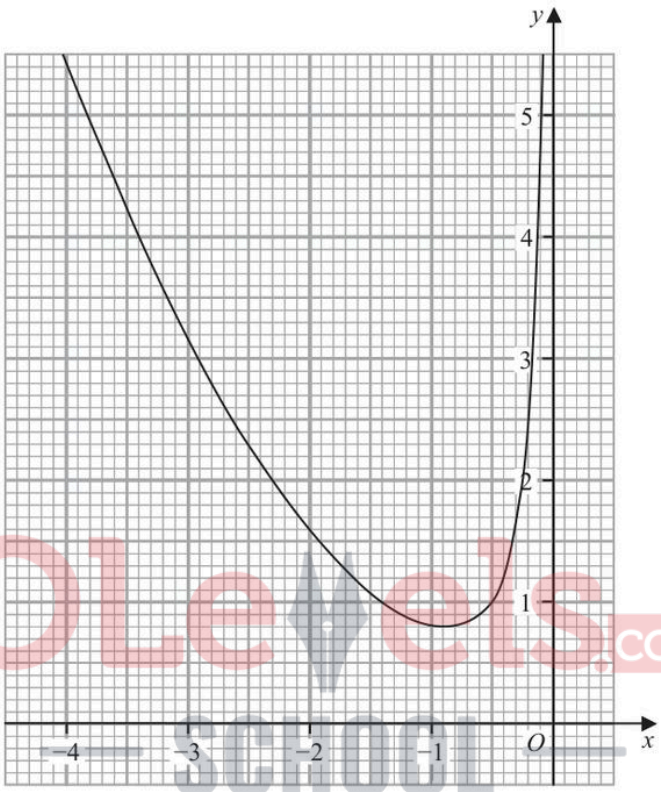


Figure 2

Figure 2 shows part of the curve with equation $y = \frac{x^2}{3} - \frac{1}{2x}$ for $-4 < x < 0$

By drawing a suitable straight line on the grid, obtain estimates, to one decimal place, of the roots of the equation $4x^3 + 3x^2 - 36x - 6 = 0$ in the interval $-4 < x < 0$

(4)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

8

P 7 6 5 0 8 A 0 8 3 2

11. Question 4 · part whole · 4 marks

2024 JUN · Paper 02

Further practice · Attempt part whole only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 4 continued

OLevels.com SCHOOL

(Total for Question 4 is 4 marks)

11. Question 4 · part whole · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

12. Question 9 · part c · 5 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

9 (a) Complete the table of values for

$$y = \frac{x^3 + 4}{5 - x}$$

giving your answers to 2 decimal places.

x	−2	−1.5	−1	−0.5	0	0.5	1	1.5	2
y	−0.57	0.10	0.5		0.8		1.25		4

(2)

(b) On the grid opposite, draw the graph of

$$y = \frac{x^3 + 4}{5 - x} \quad \text{for } -2 \leqslant x \leqslant 2$$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation $x^3 - x^2 + 8x - 11 = 0$ in the interval $-2 \leqslant x \leqslant 2$

(5)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

24

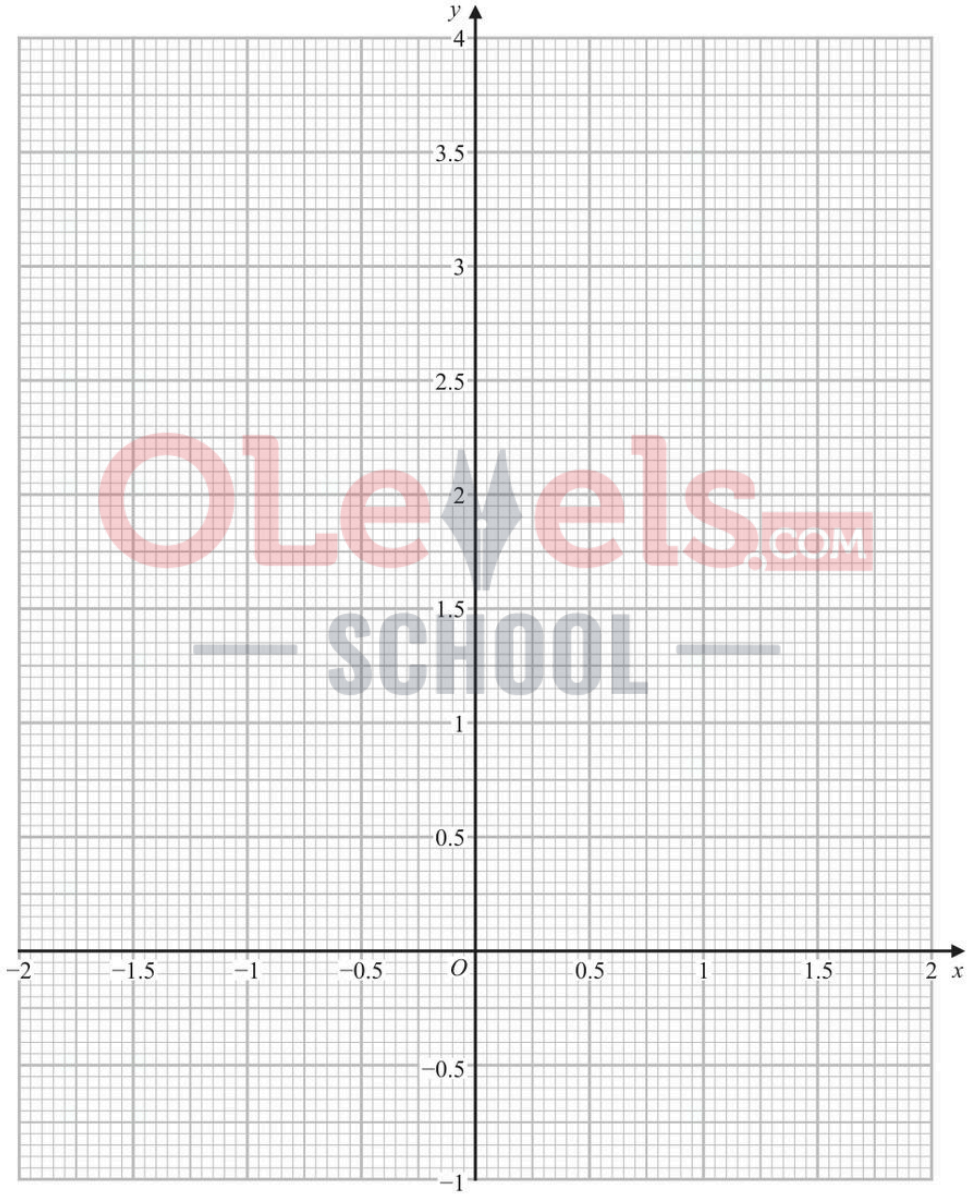
P 6 6 3 0 9 A 0 2 4 3 6

12. Question 9 · part c · 5 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 9 continued



Turn over for a spare grid if you need to redraw your graph.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

12. Question 9 · part c · 5 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 9 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

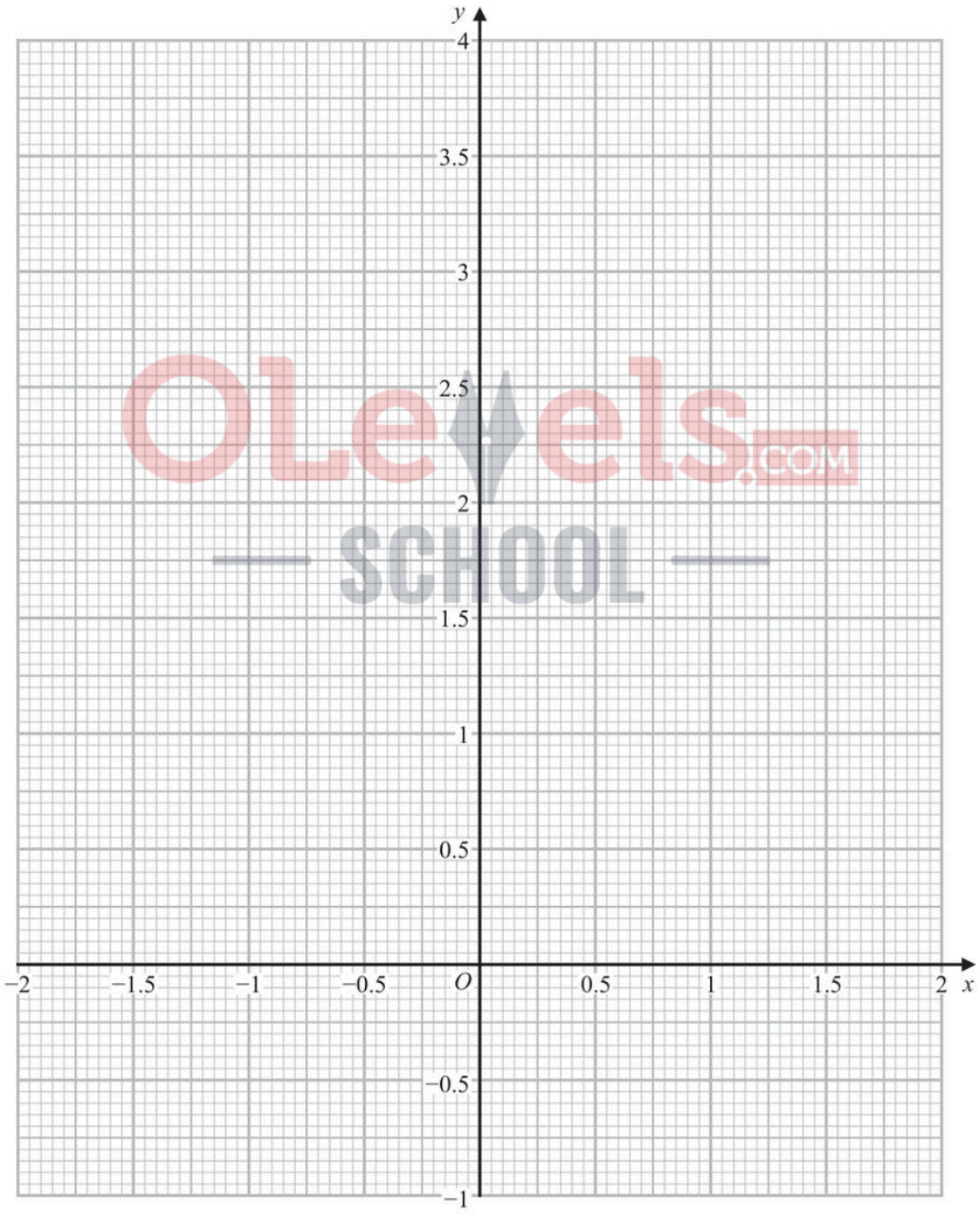
12. Question 9 · part c · 5 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 9 continued

Only use this grid if you need to redraw your graph.



OLevels.com SCHOOL

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(Total for Question 9 is 9 marks)

12. Question 9 · part c · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL